



# R for Data Science Project – Team Plan (AMR)

## Project Title

**A Statistical Analysis of Antimicrobial Resistance (AMR) Trends in Europe Using Surveillance Data**

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## Project Overview

Antimicrobial Resistance (AMR) is a major global healthcare challenge where bacteria become resistant to commonly used antibiotics. This project focuses on the **statistical analysis of AMR patterns** using publicly available European surveillance data.

The objective is **not to build prediction or machine learning models**, but to **analyze resistance distribution, trends, and statistically significant differences** using the R programming language.

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## What We Are Doing (IMPORTANT)

This project is a **statistics-based data analysis project**, aligned with the R for Data Science syllabus.

We will perform:

### **1** Descriptive Statistical Analysis

- Resistance vs susceptibility counts
- Resistance percentages
- Year-wise summaries

### **2** Trend Analysis

- How antibiotic resistance changes over years
- Identify increasing or decreasing resistance trends

### **3 Comparative Analysis**

- Compare resistance across different years
- Compare resistance across antibiotic groups

### **4 Hypothesis Testing**

- Test whether resistance differs significantly across years or antibiotics
- Use statistical tests such as:
  - Chi-square test
  - ANOVA (if applicable)

### **5 Data Visualization**

- Line charts (trends)
  - Bar charts (resistance distribution)
  - Heatmaps (optional)
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## **Dataset Used**

- European Antimicrobial Resistance Surveillance Data
  - Aggregated data over multiple years
  - Data includes:
    - Bacteria type
    - Antibiotic class
    - Resistance percentages
    - Year-wise information
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# Tools & Language

- R programming language
- Base R functions
- Simple statistical tests
- Standard R plots

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## Team Work Division (Suggested)

| Team Member     | Responsibility                       |
|-----------------|--------------------------------------|
| Member 1 (Lead) | Overall coordination, final analysis |
| Member 2        | Descriptive statistics               |
| Member 3        | Hypothesis testing                   |
| Member 4        | Visualization & report formatting    |

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## Final Outcome

- A complete statistical analysis report
- Clear interpretation of AMR trends
- Proper use of R for data analysis
- Well-structured academic submission